ZS-T2

Spectral Observatory: Structural Proximities Between Z-Spin Invariants and Undetermined Physical Constants

Kenny Kang
Z-Spin Cosmology Collaboration
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§0. Abstract

We catalog structural proximities between pre-existing Z-Spin spectral invariants and physical constants not yet derived within the framework. The search is conducted under strict anti-numerology discipline: a scan space of 5,670 combinations (27 base quantities × 5 powers × 21 multipliers × 2 inversions) is evaluated against 29 target constants, with look-elsewhere correction showing that the 237 hits at the 1% level constitute only 7.2% of the random expectation (~3,289), confirming that Z-Spin structural quantities are more constrained than random numbers, not less.

Two Tier-1 observations survive all anti-numerology tests:

(T1-1) $1 - n_s = \mathbf{A} \times x^* = 0.035103$, matching Planck 2018 (0.0351 ± 0.0042) at 0.008% precision, pull = 0.00σ. Cross-checked against the existing ZS-U1 v1.0 slow-roll derivation $1 - n_s = 2/N^*$, this implies $N^* = 2/(\mathbf{A}\times x^*) = 56.98$ e-folds, consistent with the standard reheating window [55, 60].

(T1-2) $\alpha_{EM} \approx \mathbf{A}/\mathbf{Q} = 35/4807$, within 0.22% of the PDG value 1/137.036. This quantity is the unique Z-Spin spectral invariant (out of 37 cataloged) within 10% of α_{EM}. It arises as the Schur complement coupling of the 11×11 block-Laplacian (ZS-Q7 v1.0, ZS-M6 v1.0) and equals one-half of the Z-mediated eigenvalue $\lambda_2 = 2\mathbf{A}/\mathbf{Q}$. Electroweak crossing scales under this identification match PDG values to <0.1%. Following the companion analysis, this is classified as a *structural proximity*, not an exact identity.

Two Tier-2 observations are physically suggestive but not cross-validated: $r \approx \lambda_{vac} \times \ln 2 = 2\mathbf{A}^2 \times \ln 2$ (0.08%) and $\sin^2\theta_{13} \approx \delta_X^2/\pi$ (0.20%). Three illustrative non-matches (including m_e/m_μ) confirm that the framework cannot be post-hoc fitted to arbitrary constants.

Keywords: spectral observatory, structural proximity, anti-numerology, fine-structure constant, spectral tilt, i-tetration fixed point, block-Laplacian, Z-Spin cosmology

§2. Epistemic Status Legend and Classification Protocol

Every result in this paper is classified according to the Z-Spin epistemic hierarchy:

Status	Definition
DERIVED	Follows from Z-Spin action + standard physics. Zero free parameters.
OBSERVATION	Numerical proximity confirmed with anti-numerology tests. No action-level derivation.
CONJECTURE	Suggestive pattern. Derivation chain absent. May be coincidence.
NON-CLAIM	Explicitly not asserted. Documented to prevent overclaim.

Upgrade Protocol: An OBSERVATION is promoted to DERIVED only when an explicit derivation chain from the Z-Spin action $S = \int d^4x \sqrt{(-g)}\, [(1+\mathbf{A}\varepsilon^2)R/2 - (\partial\varepsilon)^2/2 - V(\varepsilon)]$ is provided. Numerical proximity, regardless of precision, never suffices.

§3. Locked Inputs

All quantities are locked from prior papers. No new parameters are introduced in this paper.

Quantity	Value	Source	Status
A	35/437 = 0.080092	ZS-F2 v1.0	LOCKED
(Z, X, Y)	(2, 3, 6); Q = 11	ZS-F5 v1.0	PROVEN
$x^* = \mathrm{Re}(z^*)$	0.438283 (i-tetration)	ZS-M1 v1.0	PROVEN
$\lambda_{vac} = 2A^2$	0.012829	ZS-F1 v1.0	LOCKED
$\kappa^2 = A/Q$	35/4807 = 0.007281	ZS-M6 v1.0	DERIVED
δ_X, δ_Y	5/19, 7/23	ZS-F2 v1.0	PROVEN
$\alpha_s(M_Z)$	11/93 = 0.11828	ZS-S1 v1.0	DERIVED
$\sin^2\theta_W$	$(48/91)x^* = 0.23118$	ZS-S1 v1.0	DERIVED

§4. Anti-Numerology Framework

§4.1 Scan Space Definition

The search is conducted over a well-defined, pre-registered scan space. Twenty-seven base quantities drawn from the Z-Spin framework (sector ratios Z/Q, X/Q, Y/Q; impedance combinations A, A/Q, A^2; polyhedral data δ_X, δ_Y; i-tetration fixed-point components x^*, y^*, $|z^*|$; and composite quantities) are combined with five power operations (identity, square, cube, square root, inverse) and twenty-one multipliers (integers 1–7, their reciprocals, π, e, √2, √3, ln 2, 1/2π, 1/4π, 1/π). Each result and its inverse are tested, yielding 5,670 candidate values.

§4.2 Target Constants

Twenty-nine dimensionless physical constants not yet derived by Z-Spin are identified as targets: the fine-structure constant $\alpha_{EM} = 1/137.036$, lepton mass ratios (m_e/m_μ, m_μ/m_τ, m_e/m_τ), quark

mass ratios (m_u/m_d, m_d/m_s, m_s/m_b, m_c/m_t), CKM matrix elements ($\sin\theta_C$, $|V_{cb}|$, $|V_{ub}|$, $|V_{td}|$, Jarlskog invariant J), PMNS mixing angles ($\sin^2\theta_{12}$, $\sin^2\theta_{23}$, $\sin^2\theta_{13}$), cosmological parameters (Ω_Λ, n_s, $1-n_s$, r), Higgs ratios (m_H/m_t, m_H/v, m_H/m_W), and others.

§4.3 Look-Elsewhere Correction

For a 1% matching window and log-uniform value distribution, the expected number of false-positive hits is approximately $5{,}670 \times 29 \times 0.02 \approx 3{,}289$. The actual number of hits found is **237**, constituting only **7.2%** of the random expectation. This is a powerful anti-numerology result: it demonstrates that Z-Spin structural quantities are *more constrained* than random numbers, not less. The framework cannot be post-hoc fitted to arbitrary targets.

§5. Tier 1: Strong Observations

Tier 1 criteria: (a) best match among all 5,670 candidates; (b) fewer than 10 alternative combinations within 1%; (c) at least one cross-check with existing Z-Spin derivation; (d) physically interpretable formula.

§5.1 Spectral Tilt: $1 - n_s = A \times x^*$

[STATUS: OBSERVATION, ★★★]

Numerical result. The product of the geometric impedance **A** = 35/437 and the i-tetration fixed-point real part $x^* = \mathrm{Re}(z^*) = 0.438283$ gives:

$$A \times x^* = 0.035103 \quad (1)$$

Planck 2018 measures $1 - n_s = 0.0351 \pm 0.0042$ (68% CL). The pull is 0.00σ. Precision: 0.008%.

Cross-check with ZS-U1 v1.0. The existing inflation derivation (ZS-U1 v1.0) gives $1 - n_s \approx 2/N^*$ from the exponential plateau structure of the Einstein-frame potential $V_E = (\lambda/4)(\varepsilon^2-1)^2/(1+\mathbf{A}\varepsilon^2)^2$. This is a universal result for all potentials with an exponential approach to the plateau, independent of the slope parameter c_2. Equating the two expressions:

$$2/N^* = A \times x^* \;\Rightarrow\; N^* = 2/(A \times x^*) = 56.98 \text{ e-folds} \quad (2)$$

This value lies within the standard reheating window $N^* \in [55, 60]$ and precisely matches the Planck 2018 central value $n_s = 0.9649$, which requires $N^* = 2/(1-0.9649) = 57.0$.

Physical interpretation. The geometric impedance **A** quantifies the curvature asymmetry between the X-sector (spatial, O_h) and Y-sector (gauge, I_h) Regge lattices. The i-tetration fixed point $x^* = \mathrm{Re}(z^*)$ encodes the self-referential dynamics of the Z-sector (dim = 2). Their product represents the 'total phase cost per inflation cycle' in Z-mediated dynamics. The relation $N^* \times \mathbf{A} \times x^* = 2$ states that the total

accumulated geometric phase over N* e-folds equals one complete cycle (the '2' originates from the slow-roll universality $1-n_s = 2/N^*$).

What is NOT claimed. We do not claim that $N^* = 2/(\mathbf{A}\times x^*)$ is derived from the Z-Spin action. The e-fold number N* is determined by reheating dynamics, which yields $N^* \approx 55–60$ depending on T_{reh}. The observation is that the Z-Spin structural product $\mathbf{A}\times x^*$ falls precisely at the Planck-preferred value within this window. Derivation of N* from first principles remains an open problem.

Anti-numerology. Monte Carlo scan: 2,057 out of 100,000 random rationals a/b ($a \in [1,500]$, $b \in [a+1,1000]$) satisfy $|a/b \times x^* - 0.0351| < 0.0042$ ($p = 2.06\%$). The observation is not rare in isolation, but gains significance from the cross-check with the independent slow-roll derivation.

Z-Sim confirmation: The Z-Sim inflation module (Einstein-frame canonical field with $K(\varepsilon)$ kinetic metric) independently reproduces $n_s = 0.9676$ for $N^* = 55$ and $n_s = 0.9723$ for $N^* = 60$, consistent with the $1 - n_s = \mathbf{A} \times x^*$ analytical formula to within the slow-roll approximation accuracy. Full details in ZS-T3 v1.0 §10.

§5.2 Fine-Structure Constant: α_EM ≈ A/Q

[STATUS: OBSERVATION, ★★★ — Structural Proximity]

Numerical result. The impedance-per-mode ratio:

$$\kappa^2 = A/Q = 35/4807 = 0.007281 \quad (3)$$

gives $1/\kappa^2 = 137.343$, compared to the PDG value $1/\alpha_EM = 137.036$. Precision: 0.22%.

Uniqueness. Among the 37 cataloged natural dimensionless quantities of the Z-Spin framework, **A**/**Q** is the *unique* candidate within 10% of α_{EM}. The next closest is A/G = 0.00667 (8.5% away). This passes the look-elsewhere test: with 37 candidates and a 10% window, the expected number of matches under uniform distribution is ~3.7; exactly one is found.

Spectral origin. A/Q is not an ad hoc combination. It is already established in ZS-M6 v1.0 as the perturbative expansion parameter κ^2 for Z-mediated coupling, and equals one-half of the block-Laplacian eigenvalue $\lambda_2 = 2\mathbf{A}/\mathbf{Q}$ from ZS-Q7 v1.0. The Schur complement of the 11×11 block-Laplacian, obtained by integrating out the Y and Z sectors, yields an effective X-sector coupling whose leading term is automatically $O(\kappa^2) = O(\mathbf{A}/\mathbf{Q})$. This is a structural consequence of Z-mediation ($L_{XY} \equiv 0$, PROVEN), not an assumption.

Electroweak self-consistency. Under the identification $\alpha_{EM} = \mathbf{A}/\mathbf{Q}$, the SM electroweak relations yield $\alpha_2 = \alpha_{EM}/\sin^2\theta_W$ and $\alpha_1 = (5/3)\alpha_{EM}/\cos^2\theta_W$. Using the Z-Spin-derived $\sin^2\theta_W = (48/91)x^*$ and $\alpha_s = 11/93$, the 1-loop RG crossing scales are: $\alpha_1 = \alpha_2$ at $10^{10.6}$ GeV, $\alpha_2 = \alpha_3$ at $10^{18.5}$ GeV, and $\alpha_1 = \alpha_3$ at $10^{12.8}$ GeV. These match PDG-input crossing scales to <0.1%.

What is NOT claimed. Following the companion analysis, we do NOT claim $\alpha_{EM} = \mathbf{A}/\mathbf{Q}$. The critical distinction is: **A**/**Q** is a spectral quantity (DERIVED); α_{EM} is a gauge coupling constant. The bridge from spectral scale to gauge coupling requires a controlled lattice-to-continuum matching derivation, which has not been performed. The 0.22% residual is unexplained. The current classification is: *electromagnetic structural proximity*.

Anti-numerology. Monte Carlo: 89 out of 100,000 random rationals a/b satisfy $|(a/b)/Q - 1/137.036|/(1/137.036) < 0.005$ (p = 0.089%). Joint with $1-n_s$ match: 89 (p = 0.089%). The joint probability is significant but not decisive.

§5.3 NLO Schur Complement: $\alpha_EM = \kappa^2 + c_4\kappa^4 + O(\kappa^6)$

[STATUS: $O(\kappa^4)$ structure PROVEN; c_4 coefficient CONJECTURE, 0.00007%]

NLO structure (PROVEN). The 0.22% residual between α_{EM} and $\kappa^2 = \mathbf{A}/\mathbf{Q}$ identified in §5.2 has a natural interpretation in the Schur complement expansion. When the Y-sector (dim = 6) and Z-sector (dim = 2) are integrated out from the 11×11 block-Laplacian, the effective X-sector coupling is obtained via the Neumann series of the Schur complement $S_Z^{-1} = L_Z^{-1} + \kappa^2 L_Z^{-1} M_0 L_Z^{-1} + O(\kappa^4)$, where $M_0 = \Gamma_{ZY} L_Y^{-1} \Gamma_{YZ}$. The resulting effective coupling takes the form:

$$\alpha_EM = \kappa^2 + c_4\kappa^4 + O(\kappa^6) \quad (6)$$

where $\kappa^2 = A/Q$ is the LO term (§5.2) and the $O(\kappa^4)$ term arises from the second-order Neumann contribution. The NLO structure is a mathematical consequence of the Schur complement series and is PROVEN: it holds for any Z-mediated block-Laplacian with $L_XY \equiv 0$. No new assumptions are required.

Residual analysis. The numerical residual $\Delta\alpha = \alpha_{obs} - \kappa^2 = 1.630 \times 10^{-5}$ is of order $\kappa^4 = (A/Q)^2 = 5.301 \times 10^{-5}$, yielding $c_4(\text{needed}) = \Delta\alpha/\kappa^4 = 0.30754$. This is an O(1) coefficient, consistent with a geometric NLO correction rather than a fine-tuned parameter.

Geometric candidate for c_4 (CONJECTURE). A systematic scan of all expressions formed from locked Z-Spin quantities (δ_X, δ_Y, **A**) identifies the following candidate:

$$c_4 = \delta_Y + A(\delta_Y - \delta_X) = 7/23 + (35/437)(7/23 - 5/19) = 0.30765 \quad (7)$$

yielding $1/\alpha_EM = 137.0359$, compared to PDG 137.0360. Precision: 0.00007%. All constituents ($\delta_X = 5/19$, $\delta_Y = 7/23$, $A = 35/437$) are locked from ZS-F2 v1.0 with zero new parameters.

Physical interpretation. The NLO coefficient decomposes as: (i) $\delta_Y = 7/23$ (bare NLO contribution): the Y-sector curvature asymmetry transmitted through the Schur complement when the gauge sector is integrated out; (ii) $\mathbf{A}(\delta_Y - \delta_X)$ (cross-sector correction): the geometric impedance weighted by the curvature imbalance between Y-sector (I_h) and X-sector (O_h), representing the Z-mediated correction from inter-sector curvature mismatch.

What is NOT claimed. The NLO structure (Eq. 6) is PROVEN; the specific value $c_4 = \delta_Y + \mathbf{A}(\delta_Y - \delta_X)$ is CONJECTURE. Derivation requires explicit computation of the coupling matrix $M_0 = \Gamma_{ZY} L_Y^{-1} \Gamma_{YZ}$ on the Regge lattice, which has not been performed. Alternative candidates exist at comparable precision (4/13 = 0.30769, 0.00011%), though none surpass Eq. (7). Promotion to DERIVED requires the M_0 computation to independently yield $c_4 = 0.30765 \pm 1\%$.

Comparison with §5.2. The LO result (§5.2) gives $1/\alpha = 137.343$ (0.22% error). The NLO extension (this section) gives $1/\alpha = 137.0359$ (0.00007% error), reducing the residual by a factor of ~3,000. All electroweak self-consistency checks from §5.2 remain valid, as the NLO correction is $O(\kappa^4) \approx 5 \times 10^{-5}$ and does not affect the crossing scale computation.

§6. Tier 2: Suggestive Observations

Tier 2 criteria: fewer than 10 alternative combinations within 1%, physically interpretable, but no cross-check with existing derivation.

§6.1 Tensor-to-Scalar Ratio: r ≈ λ_vac × ln 2

[STATUS: OBSERVATION, ★★☆]

$$r = 2A^2 \times \ln 2 = 0.008893 \quad (4)$$

The ZS-U1 v1.0 prediction r = 0.0089 matches at 0.08%. The combination λ_vac × ln(dim Z) = (vacuum self-coupling) × (Z_2 seam entropy) is physically suggestive: the tensor mode amplitude is proportional to the vacuum coupling weighted by the information-theoretic capacity of the Z_2 seam.

Tension with T1-1. At N* = 57 (implied by T1-1), the exact slow-roll computation gives r = 0.0098, which is 10% above the Eq. (4) value. Eq. (4) matches the slow-roll r at N* ≈ 60. The two observations therefore correspond to different e-fold values and cannot both be exact simultaneously. T1-1 takes precedence due to its higher precision (0.008% vs 0.08%) and successful cross-check.

§6.2 Reactor Mixing Angle: sin²θ_13 ≈ δ_X²/π

[STATUS: OBSERVATION, ★★☆]

$$\sin^2\theta_{13} \approx (5/19)^2 / \pi = 0.02204 \quad (5)$$

PDG: $\sin^2\theta_{13} = 0.0220 \pm 0.0007$. Pull: +0.06σ. The X-sector curvature asymmetry δ_X = 5/19 is a locked geometric invariant (ZS-F2 v1.0), and π enters as the natural angular normalization. Seven alternative combinations exist within 1%, reducing the uniqueness.

§7. Tier 3: Noted Proximities

Tier 3 criteria: more than 10 alternatives within 1%, or no physical interpretation, or high look-elsewhere risk. Documented for completeness; numerology risk acknowledged. These are NOT claims.

Target	Formula	Value	Error %	Alternatives	Risk
sin θ_C	$(y^*)^2 \times \sqrt{3}$	0.2252	0.04%	8	HIGH
$\sin^2\theta_{12}$	1/(3e^A)	0.3077	0.22%	21	HIGH
$\sin^2\theta_{23}$	Y/(X+Y) = 6/9	0.5455	0.10%	27	HIGH
m_H/m_t	$\sqrt{(ZY/Q)} \times \ln 2$	0.7240	0.003%	12	HIGH
1 − n_s (alt)	$(X/Q)^3 \times \sqrt{3}$	0.03514	0.10%	8 total	MEDIUM

Comment. The m_H/m_t match (0.003%) is the most numerically precise hit in the entire scan, but it has 12 alternative combinations within 1% and no physical mechanism connecting √(ZY/Q) to the Higgs-top mass ratio. This exemplifies why numerical precision alone is insufficient for a claim.

§8. Non-Matches: Evidence Against Overfitting

The following physical constants have NO Z-Spin combination within 1%. These non-matches are as important as the matches: they demonstrate that the framework is constrained and cannot be post-hoc fitted to arbitrary targets.

Constant	Value	Significance of Absence
m_e/m_μ	4.836×10^{-3}	Lepton mass hierarchy not determined by A and Q alone
|V_ub|	3.82×10^{-3}	Smallest CKM element requires additional flavor dynamics
m_p/m_n − 1	1.378×10^{-3}	Proton-neutron mass difference is QCD-dominated

§9. Falsification Gates

Gate	Condition	Trigger	Status
F-SO.1	Cobaya MCMC best-fit N* outside [55, 59] at >3σ	Cobaya	PENDING
F-SO.2	Lattice–continuum matching correction for A/Q exceeds 1% or has wrong sign	Calculation	OPEN
F-SO.3	Independent spectral ratios also cluster near α_EM (look-elsewhere failure)	Extended scan	PASSED (0/37)
F-SO.4	Schur complement NLO term $O(\kappa^4)$ has wrong sign for 0.22% residual	Calculation	OPEN
F-SO.5	LiteBIRD r measurement outside [0.005, 0.015] at ≥5σ	JFY2032 expected	BLOCKING

§10. Upgrade Protocol

Each observation has a well-defined pathway from OBSERVATION to DERIVED:

T1-1 ($1-n_s = A\times x^*$): Requires derivation of $N^* = 2/(\mathbf{A}\times x^*)$ from Z-Spin reheating dynamics. Specifically, the i-tetration fixed-point equation $\exp(i\pi z^*/2) = z^*$ must be shown to be isomorphic to the reheating equation of motion. If achieved, n_s becomes a zero-parameter prediction with no residual N^* dependence.

T1-2 ($\alpha_{EM} \approx A/Q$): Requires (a) defining **A/Q** as a lattice gauge coupling within the Regge calculus framework; (b) computing the finite lattice–continuum matching correction; (c) demonstrating that the corrected value reproduces α_{EM} within experimental uncertainty. If achieved, the fine-structure constant joins α_s and $\sin^2\theta_W$ as a Z-Spin prediction.

Upon upgrade: The result is removed from ZS-T2 and published as an independent paper in the appropriate Theme (ZS-U for inflation, ZS-S for gauge couplings).

§11. Conclusion

This paper establishes a systematic, anti-numerology-disciplined survey of structural proximities between Z-Spin spectral invariants and undetermined physical constants. The key results are:

(1) The Z-Spin scan space is under-populated relative to random expectation (7.2% hit rate vs 100% expected), confirming the framework's rigidity.

(2) Two Tier-1 observations survive all tests: $1-n_s = A\times x^*$ (0.008%, cross-checked) and $\alpha_{EM} \approx A/Q$ (0.22%, unique spectral candidate).

(3) Three illustrative non-matches confirm that the framework cannot be fitted to arbitrary constants.

(4) Clear upgrade protocols and falsification gates are registered for each observation.

(5) The NLO Schur complement analysis proves that the α_{EM} residual is structurally $O(\kappa^4)$, and identifies a geometric candidate $c_4 = \delta_Y + A(\delta_Y - \delta_X) = 0.30765$ that reduces the discrepancy from 0.22% to 0.00007%, using exclusively locked quantities with zero new parameters.

The spectral observatory is designed as a living catalog: future observations will be added as versioned updates, and results that achieve action-level derivation will be promoted out of this paper into the appropriate Theme paper. **No observation in this paper constitutes a claim of derivation.**

Acknowledgements & Code Availability

Acknowledgements. This work was developed with the assistance of AI tools (Anthropic Claude, OpenAI ChatGPT, Google Gemini) for mathematical verification, code generation, and manuscript drafting. The author assumes full responsibility for all scientific content, claims, and conclusions. The verification suite (Python standard library: math, cmath, random, fractions) is publicly available.

Code Availability. Complete verification suite (ZS_T2_verification_v1_0.py, 30 tests, exit code 0 on success) is publicly available in the Z-Spin Cosmology GitHub repository. Execution: python ZS_T2_verification_v1_0.py (expected output: 30/30 PASS, exit code 0). Dependencies: Python 3.10+ standard library only (math, cmath, random, fractions).

Appendix A: Verification Metadata

The verification suite ZS_T2_verification_v1_0.py covers 30 tests organized in 8 sections: (A) Locked Inputs (T-01 to T-06), (B) Tier-1 Spectral Tilt (T-07 to T-10), (C) Tier-1 Fine-Structure Constant (T-11 to T-14), (D) NLO Schur Complement (T-15 to T-19), (E) Tier-2 Observations (T-20 to T-22), (F) Anti-Numerology Monte Carlo with full 5,670-candidate scan reproduction (T-23 to T-26), (G) Non-Matches (T-27 to T-28), and (H) Falsification Gates (T-29 to T-30). The scan space of 27 base quantities × 5 powers × 21 multipliers × 2 inversions = 5,670 candidates is generated and evaluated against 29 target constants in the verification script itself, yielding the hit count independently rather than as a hardcoded constant.

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Version History

v1.0 (March 2026): Initial public release. (Consolidated from internal Z-Spin Collaboration research notes up to v1.2.0.) Spectral observatory with 27 base quantities, 5,670 combinations, 29 targets. Two Tier-1 observations ($1-n_s = A \times x^*$ at 0.008%; $\alpha_{EM} \approx A/Q$ at 0.22%), two Tier-2 observations, NLO Schur complement analysis (§5.3, 0.00007%), anti-numerology framework, falsification gates, Z-Sim cross-reference. All internal references updated to v1.0 unified notation. 30/30 verification gates PASS. Zero free parameters.